5.6 Compare and Explain Nets Teacher Key

Teacher use

A folds into a cube. Its middle column provides four faces around the cube; the two side squares close opposite ends. B does not: a straight row folds around the same belt, leaving ends open and causing repeated or overlapping face positions.

Each face has area $2 \times 2 = 4$ cm²; total surface area is $6 \times 4 = 24$ cm². Both flat drawings have area 24 cm², but area alone does not establish whether faces meet without gaps or overlap.

Evidence to listen for

Ask students to identify a particular overlapping or missing face. If they only count six squares, have them track the faces while folding. A student who predicts incorrectly but revises with accurate fold evidence has met the reasoning goal.